

Rouge Construction Network Topology: WAN and VPN Technology

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Rouge Construction has hopes to expand to Western Kentucky, prioritizes supply chain operations, requires leadership training for employees, and increases sales within the next 4 years. As the company expands and makes to accomplish these tasks, having branch offices, services available on construction sites, and catering to home/traveling users is inevitable. With these goals in mind and with the current plan of a wireless mesh network as the foundation, WAN and VPN technology is essential.

A dual-port WAN router was used for a branch office focused on Operations and Supply Chain Management and a branch network for a warehouse to log materials and service machinery while ensuring data is transmitted securely. Since there will be necessary training and administrative work to be done throughout the expansion and customer care a VPN connection is established based on the admin office. As seen in the diagram, a VPN gateway connects the admin office to the internet allowing for traveling employees, at home employees to work without being at the office. This allows employees to complete training, work at home, and any other necessary tasks without being forced to come into the office outside of work hours. Creating this private network provides enough service for a remote office such as a nearby construction site where tablets, laptops, or other devices may require service while on site. In using the VPN gateway, remote clients do not need to run VPN software in order to establish a connection, since this is considered a client-to-site connection.

